Financial Fraud Detection Report

1. Introduction

Financial fraud detection is an important problem in modern digital transactions. With the rapid growth of online payments, detecting fraudulent activities has become essential to prevent financial loss.

This project focuses on identifying fraudulent transactions using machine learning techniques. Various models such as Logistic Regression, Random Forest, and XGBoost were implemented to improve fraud detection accuracy.

2. Dataset Overview

The dataset contains 50,000 transaction records with multiple features related to user activity and transaction details.

Dataset Features: Transaction_ID, User_ID, Transaction_Amount, Transaction_Type, Account_Balance, Device_Type, Location, Merchant_Category, Previous_Fraudulent_Activity, Daily_Transaction_Count, Card_Type, Card_Age, Fraud_Label.

Target Variable:
 Normal Transaction is presented by 0
 Fraudulent Transaction is presented by 1

	Transaction_ID	User_ID	Transaction_Amount	Transaction_Type	Date	Account_Balance	Device_Type	Location	Merchant_Category	Previous_Fraudulent_Activity	Daily_Transaction_Count	Card_Type	Card_Age	Fraud_Label
0	TXN_33553	USER_1834	39.79	POS	14-Aug-23	93213.17	Laptop	Sydney	Travel	0	7	Amex	65	0
1	TXN_9427	USER_7875	1.19	Bank Transfer	07-Jun-23	75725.25	Mobile	New York	Clothing	0	13	Mastercard	186	1
2	TXN_199	USER_2734	28.96	Online	20-Jun-23	1588.96	Tablet	Mumbai	Restaurants	0	14	Visa	226	1
3	TXN_12447	USER_2617	254.32	ATM Withdrawal	07-Dec-23	76807.20	Tablet	New York	Clothing	0	8	Visa	76	1
4	TXN_39489	USER_2014	31.28	POS	11-Nov-23	92354.66	Mobile	Mumbai	Electronics	1	14	Mastercard	140	1

3. Data Preprocessing & Feature Engineering

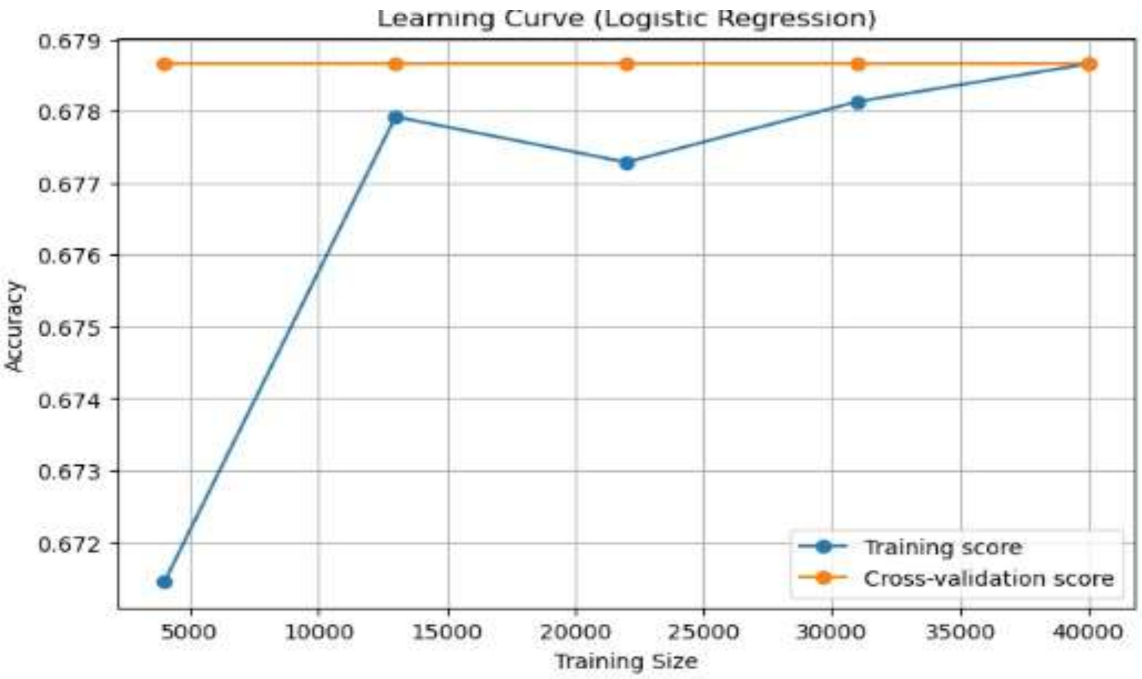
- Removed unnecessary or duplicate data.
- Converted categorical variables using encoding.
- Applied feature scaling using StandardScaler.

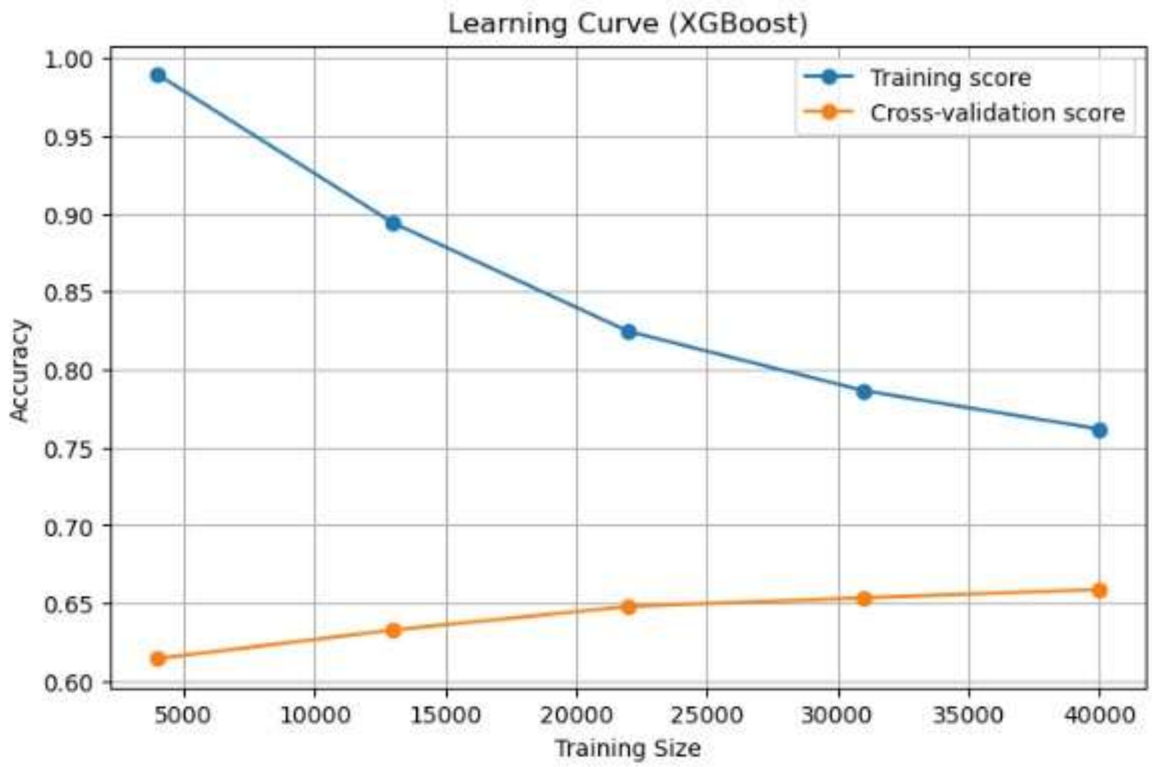
- Created new time-based features: Hour, Day, Month.
- Handled class imbalance using SMOTE (Synthetic Minority Oversampling Technique).

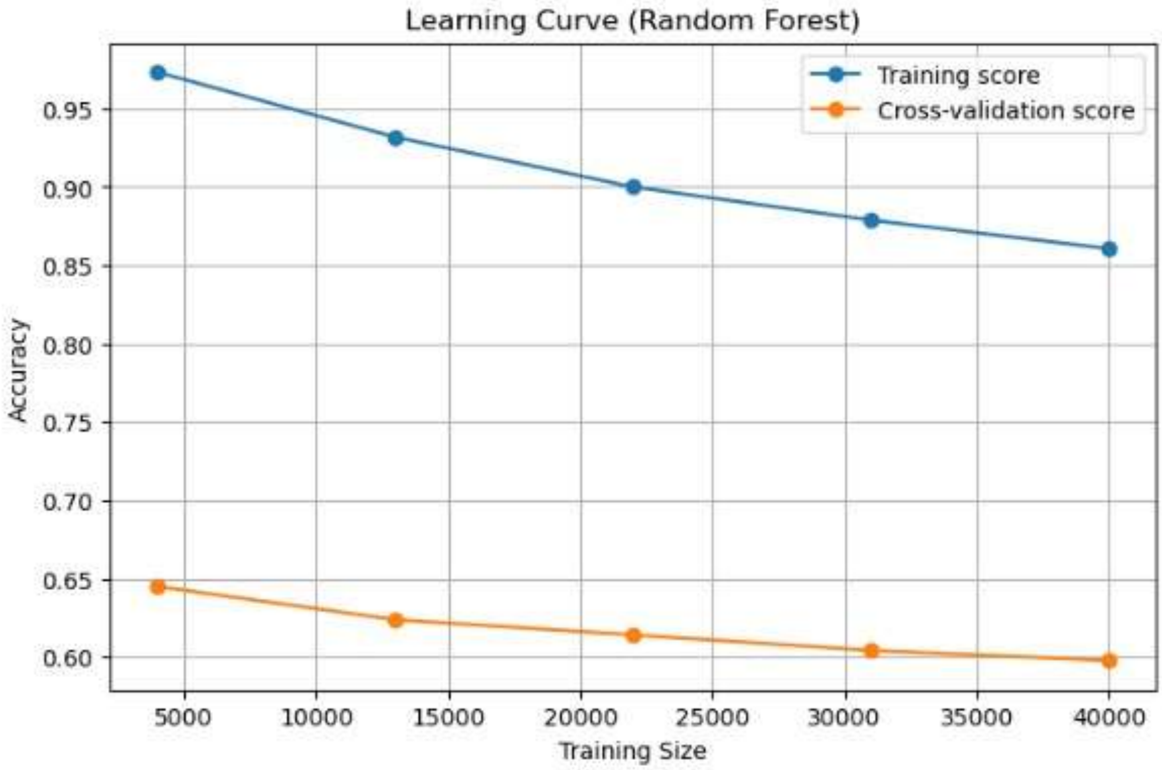
4. Machine Learning Models Used

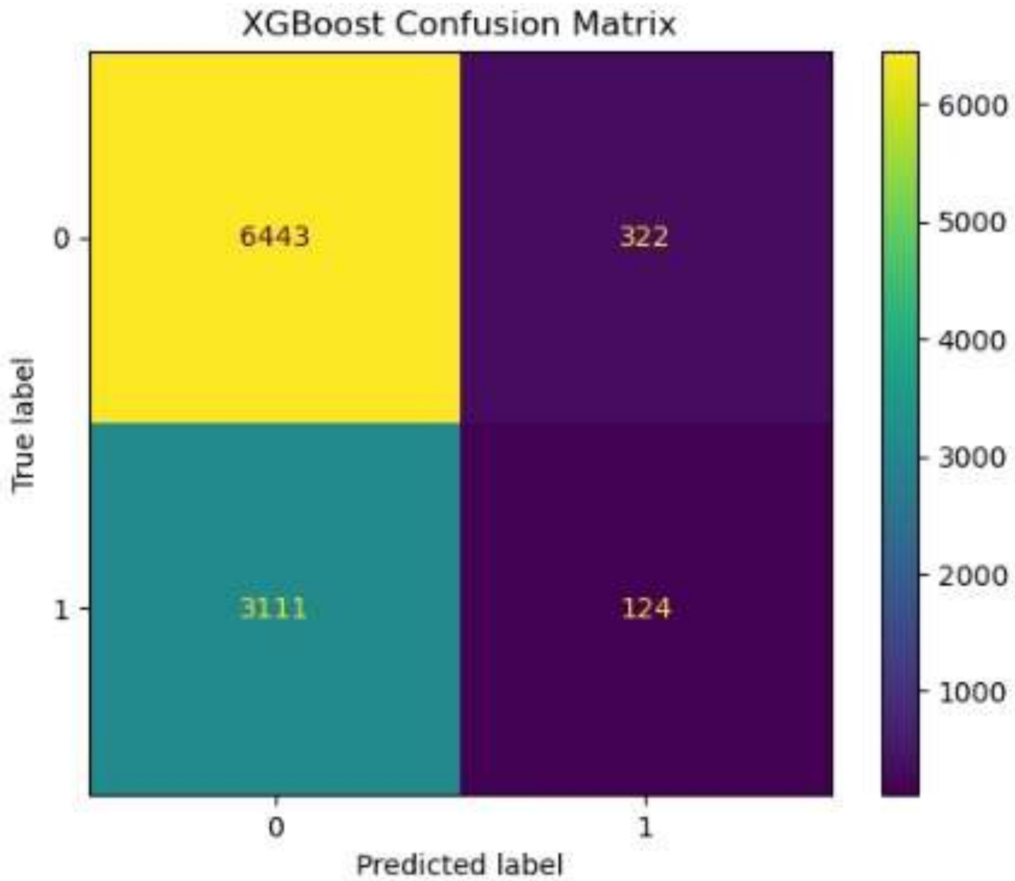
Logistic Regression
Random Forest
Classifier XGBoost
Classifier

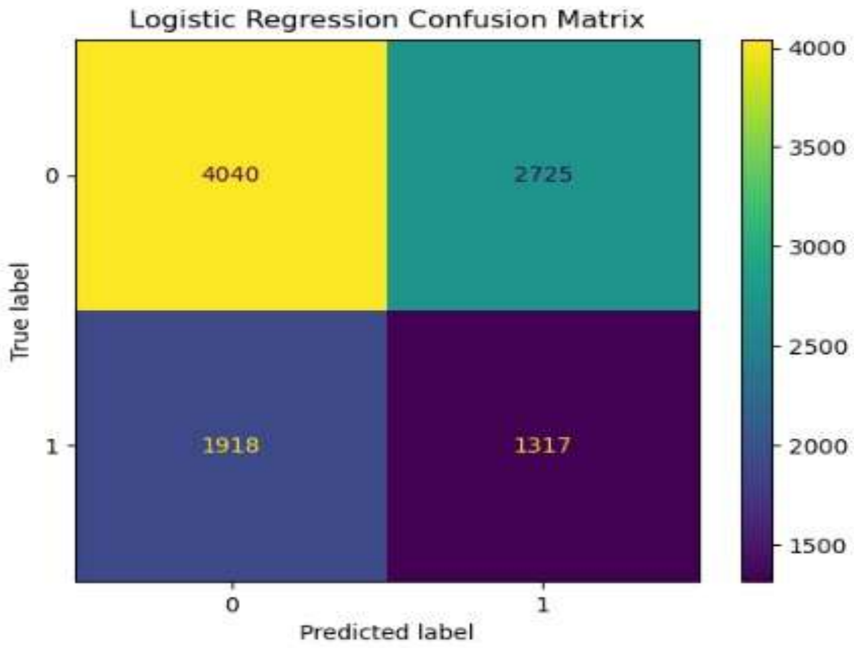
These models were used to classify transactions as fraud or non-fraud.

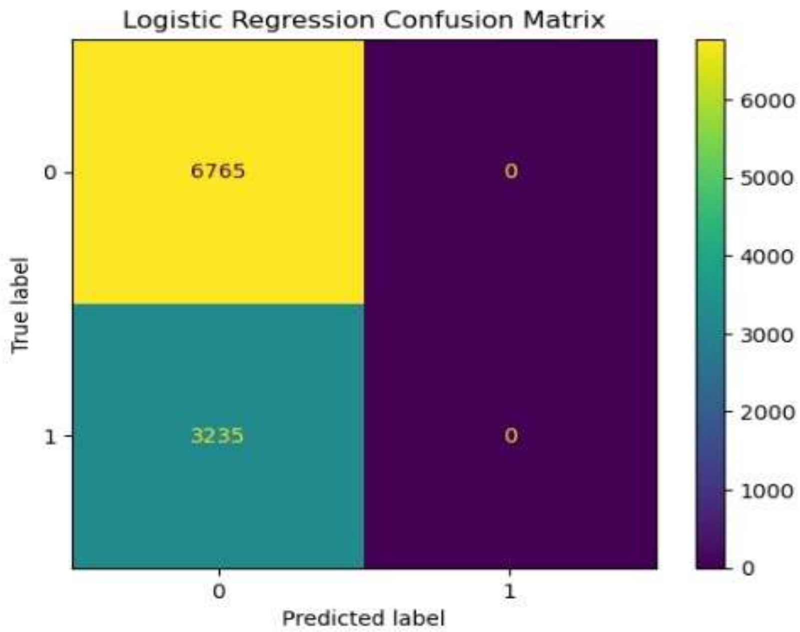


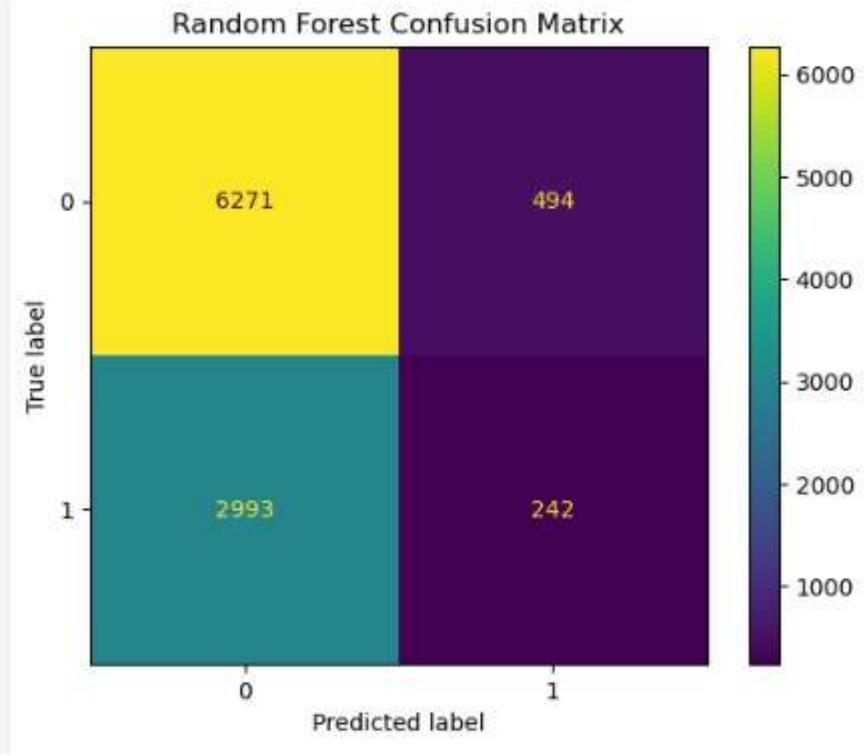



Learning Curve (Random Forest)
Training score
Cross-validation score
Accuracy
Training Size
0.95
0.90
0.85
0.80
0.75
0.70
0.65
0.60
5000
10000
15000
20000
25000
30000
35000
40000


XGBoost Confusion Matrix
6443
322
3111
124
True label
Predicted label
0
1
6000
5000
4000
3000
2000
1000


Logistic Regression Confusion Matrix
4040
2725
1918
1317
True label
Predicted label
0
1
4000
3500
3000
2500
2000
1500


Logistic Regression Confusion Matrix
6765
0
3235
0
True label
Predicted label
0
1
6000
5000
4000
3000
2000
1000
0


Random Forest Confusion Matrix
6271
494
2993
242
True label
Predicted label
0
1
6000
5000
4000
3000
2000
1000

5. Learning Curve Analysis

Learning curves were generated for each model to analyze performance.

- Logistic Regression shows stable learning but lower performance.
- Random Forest shows improved performance with reduced overfitting.
- XGBoost provides better generalization and higher accuracy.

6. Model Evaluation & Performance Metrics

Models were evaluated using Accuracy, Precision, Recall, F1-score, Confusion Matrix. Key Observations:

- Initial models had low recall for fraud detection.
- After applying SMOTE, recall improved significantly.
- Threshold tuning helped improve fraud detection sensitivity.

7. Conclusion & Future Scope

This project successfully demonstrates the use of machine learning in detecting fraudulent transactions.

- Random Forest and XGBoost performed better than Logistic Regression.
- SMOTE improved class balance.
- Threshold tuning improved fraud detection.

Future Scope:

- Use deep learning models for better accuracy.
- Implement real-time fraud detection system.
- Improve feature engineering.
- Deploy the model using web applications.